

MECHENG726: Acoustics for Engineers

Chapter 3: Science of hearing (ver.1.0)

Learning outcomes

By the end of these lecture notes, students will understand

- ▶ Physiology of ears; what our hearing system consists of and how they work to convert physical sound to auditory perception
- ▶ Function of hearing; characteristics of audible and inaudible sound
- ▶ Psychology of hearing; the relationship between objective sound and subjective sound
- ▶ Spatial hearing; how acoustic environment affects our hearing

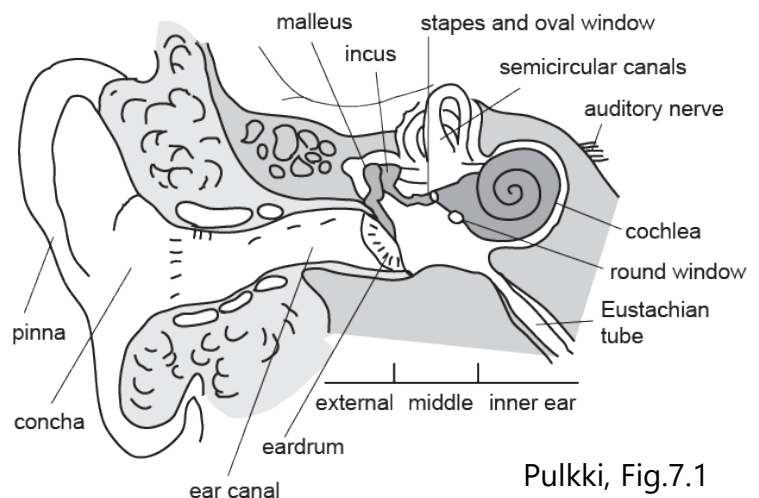
Section 1: Physiology of human hearing

3

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Global structure of the ear

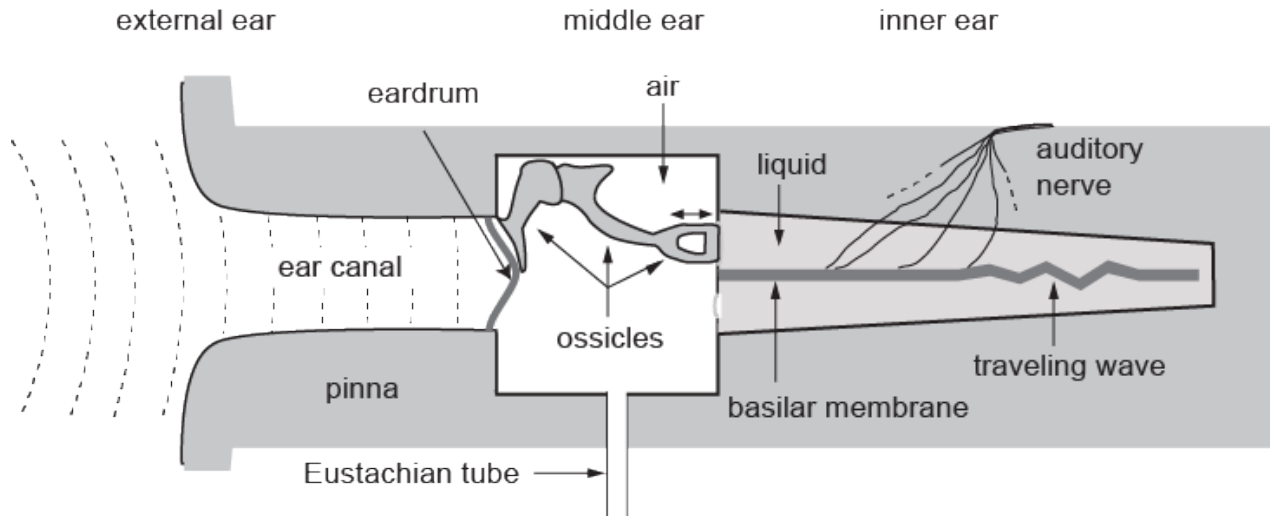
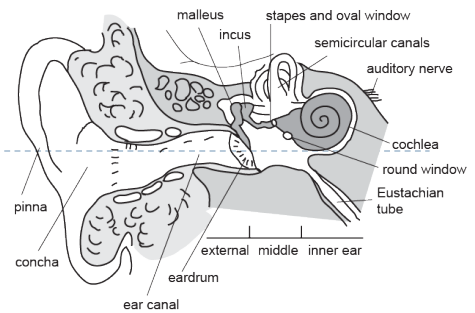
- ▶ Consist of three parts: *outer (external) ear*, *middle ear*, and *inner ear*
 - ▶ Outer ear: Captures sound waves travelling in the air, pinna – ear canal – eardrum
 - ▶ Middle ear: Mechanical conduction of vibration of eardrum to cochlea
 - ▶ Inner ear: Mechanical – to - neural transduction to send neural signal to the brain



Pulkki, Fig.7.1

Global structure of the ear

► Simplified structure of the ear

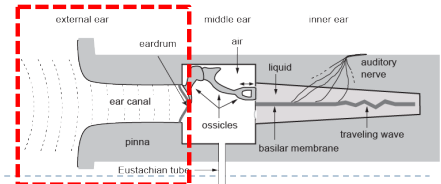


Pulkki, Fig.7.2

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Outer ear

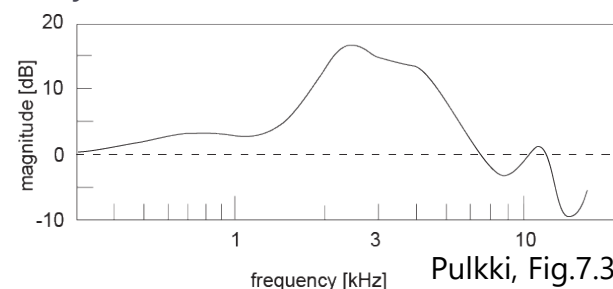


► **Passive** and **linear** system, functioning entirely based on laws of acoustics wave propagation

► Roles of parts

- **Pinna**: influences sound propagation to the ear differently depending on the source direction (i.e. front/back, top/down)
- **Ear canal**: hard-walled tube, 22.5 mm long, 7.5 mm diameter, works as $\frac{1}{4}$ wavelength resonator ($f_r \sim 3\text{-}4\text{ kHz} \ \& \ 10\text{ kHz}$)
- **Ear drum**: converts sound waves into mechanical vibrations passed to the middle ear, i.e. boundary with the middle ear

► Bandpass filter like frequency response ($f_{\text{centre}} \sim 2\text{-}3\text{ kHz}$)

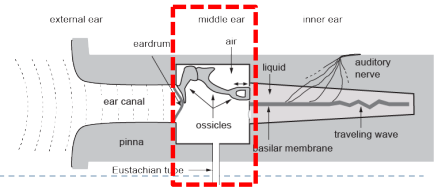


Pulkki, Fig.7.3

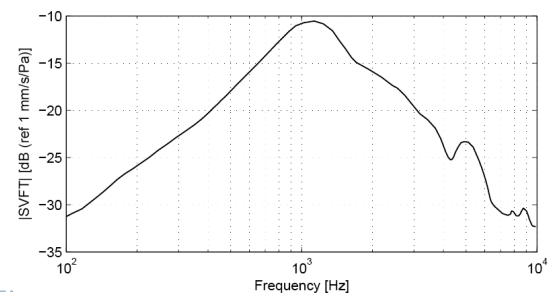
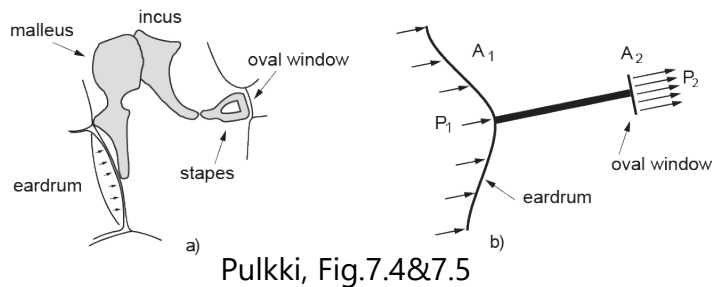
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Middle ear



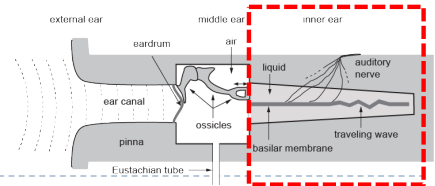
- ▶ Air-filled cavity between eardrum and inner ear
- ▶ **Transmits mechanical vibration** from eardrum to the liquid medium in cochlea (inner ear) via little bones (ossicles): impedance matching between air and liquid
- ▶ Efficient sound energy transfer realised by:
 - ▶ Lever mechanism by ossicles
 - ▶ Area ratio between eardrum and oval window
- ▶ Bandpass filter like frequency response ($f_{\text{centre}} \sim 1\text{-}2\text{ kHz}$)



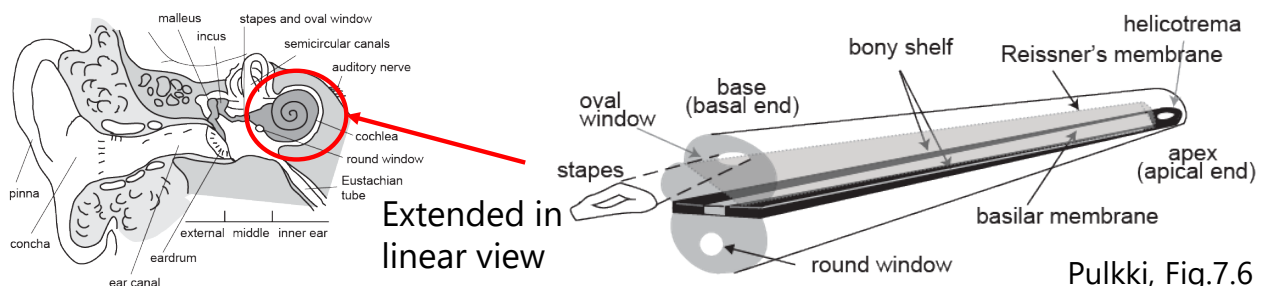
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Inner ear



- ▶ Consists of **cochlea** and semicircular canals, but only cochlea is contributing to monaural hearing
- ▶ Cochlea: spiral-shaped liquid filled tube with **basilar membrane** in the middle along its length dividing the tube into two parts; transforming mechanical vibration of sound to neural signals
- ▶ Depending on frequency of the sound, different parts of basilar membrane vibrate acting as a bandpass filter



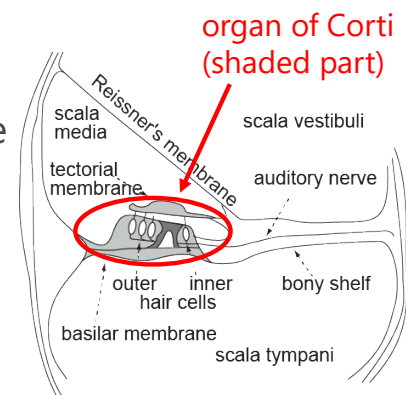
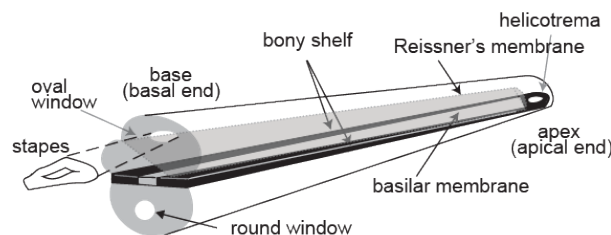
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Inner ear

- ▶ A vibration sensitive structure called *organ of Corti* resides inside the basilar membrane
- ▶ Organ of Corti has many receptor cells, *hair cells*, which converts vibration pattern of basilar membrane into neural signals transmitted to auditory nerve
- ▶ Two different types of hair cells, *inner hair cells* and *outer hair cells*

- ▶ Inner: primary receptors of vibration
- ▶ Outer: control vibration state of the membrane



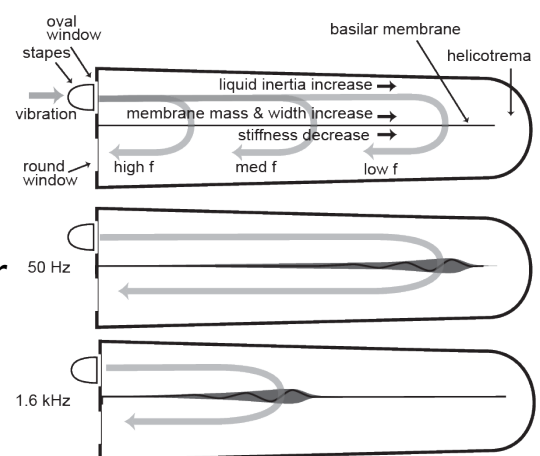
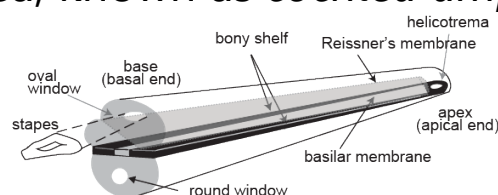
Pulkki, Fig.7.7

▶ 9

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Inner ear

- ▶ Basilar membrane gets wider, more massive, and loosely moving towards the apical end, allowing each point of the membrane resonates at different frequency
- ▶ Inner hair cells on the resonating parts transmit neural signals, allowing the brain to recognise the sound of particular frequencies
- ▶ Outer hair cells actively control the movement of the membrane to improve frequency selectivity of cochlea, known as *cochlea amplifier*



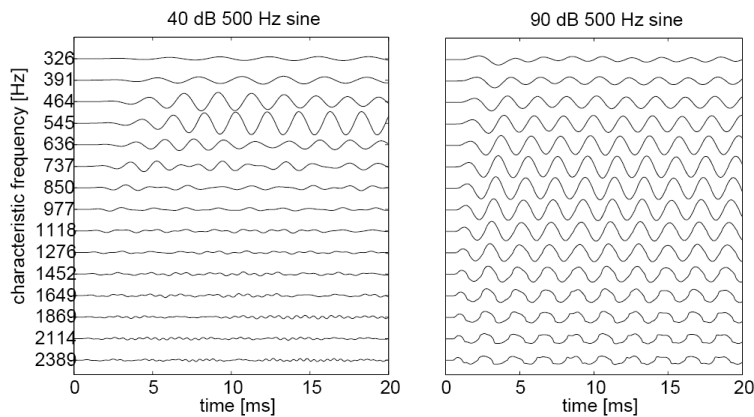
Pulkki, Fig.7.9

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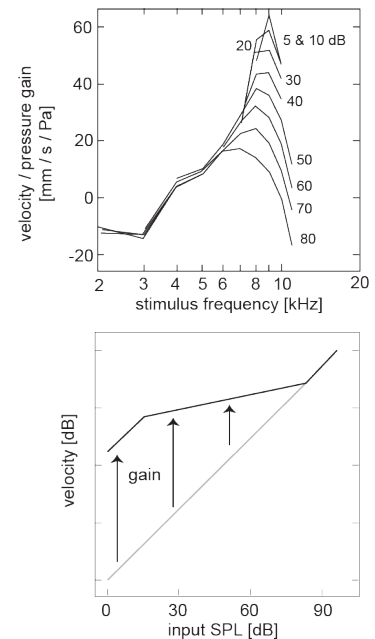
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Inner ear

- ▶ Cochlea amplifier is a non-linear feedback process
 - ▶ Weak sound are emphasised with high frequency selectivity
 - ▶ Strong sound activates wider
 - ▶ 5-10 ms delay until amplification is in effect



Pulkki, Fig.7.11&12



▶ 11

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Section 2: Basic function of hearing

Auditory object

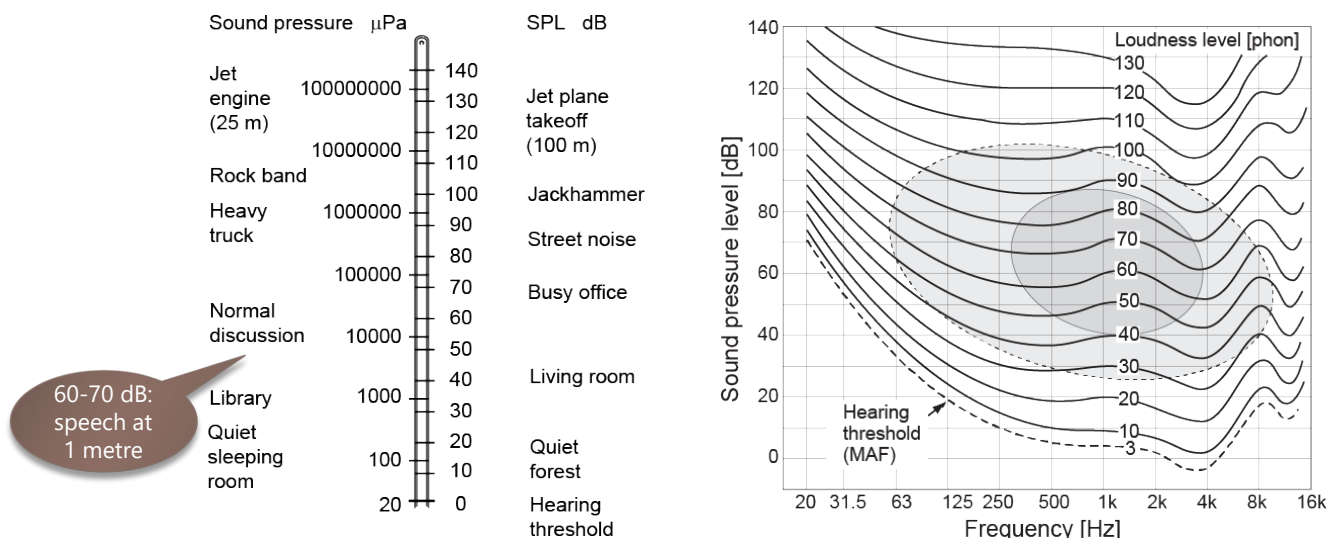
- ▶ As we learnt in Section 1, vibration of air is perceived as a sound by the sophisticated functions of the ear. There are some limits for the ear to perceive vibration as a sound.
- ▶ Two physical attributes of sound affecting its audibility:
 - ▶ **Amplitude** (= Sound pressure level (SPL))
 - ▶ Too low: Sound gets inaudible
 - ▶ Too high: Sensation of pain is added; eventually ears are damaged
 - ▶ **Frequency**
 - ▶ Too low: Not perceived as sound but felt as a sensation of vibration
 - ▶ Too high: Sensation of pain, then not perceived at all
- ▶ When multiple sounds are presented, they influence each other's audibility, known as **auditory masking**.

▶ 13

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Effective hearing area

- ▶ Working range of possible signal amplitude and frequency:
 - ▶ Amplitude: **20 μ Pa** (defined as 0 dB SPL) to **60 Pa** (~130 dB SPL)
 - ▶ Frequency: **20 Hz** to **20 kHz** (<20Hz *infrasound*; >20kHz *ultrasound*)



▶ 14

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Pulkki, Fig.9.1&9.2

Effective hearing area

- ▶ Total range used in our daily life is much smaller:

- ▶ Telephone connection: 300 – 3.4 kHz, 40 - 70 dB SPL
- ▶ Oral communication: 125 – 8 kHz, 40 – 90 dB SPL
- ▶ Acoustic music: 50 – 10 kHz, 20 – 100 dB SPL

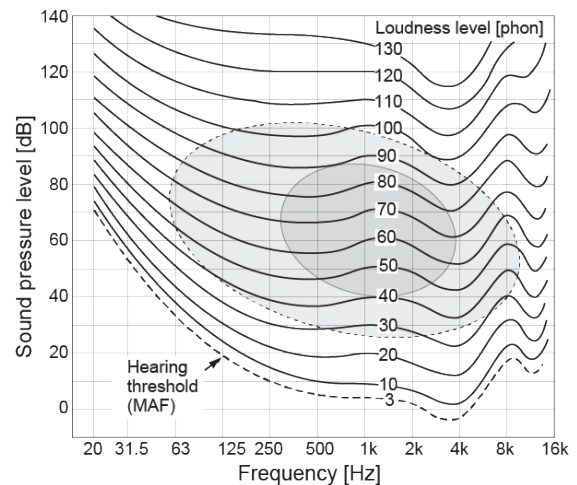
Small range



Large range

- ▶ Changing amplitude of sound affects its **loudness level*** for listeners.

- ▶ **Equal loudness curve** represents the SPL of sound to provide the same loudness level at different frequency.

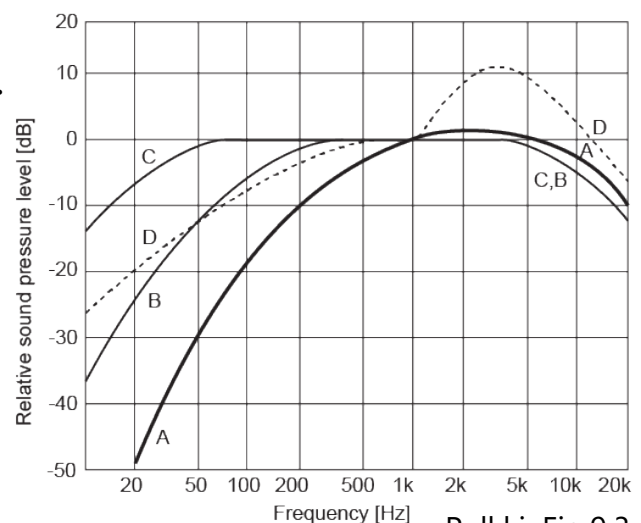


Effective hearing area

- ▶ Equal loudness curve showing varying SPL for the same loudness suggests the SPL does not match the subjective perception of sound.

- ▶ To quantify the loudness of sound, the concept of **sound level** is defined, which is the SPL weighted with frequency.

- ▶ "A-weighted" sound level, a rough estimate of inverted equal loudness curve, is commonly used*, denoted as dBA.

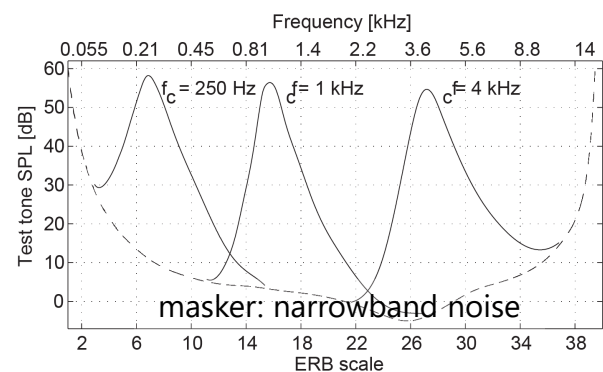
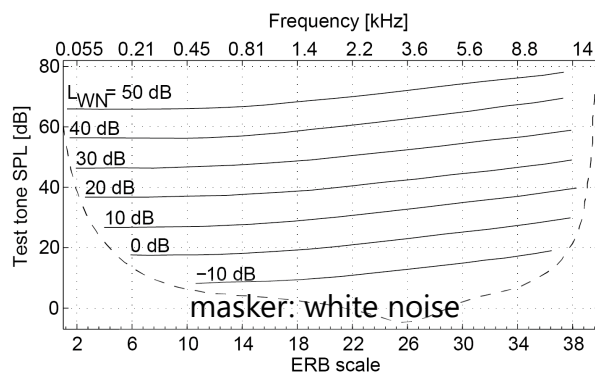


Auditory masking

- ▶ **Auditory masking** is the phenomenon where soft sounds become inaudible in the presence of louder sounds.
- ▶ Two types of auditory masking:
 - ▶ Spectral masking
 - ▶ It occurs when a sound with certain spectral content makes the detection of another sound with different spectral content harder
 - ▶ Caused by the change of threshold of hearing by the existence of other sound
 - ▶ Temporal masking
 - ▶ A sound also masks in time; affecting the audibility of a preceding (*backward masking*) or following (*forward masking*) sound
 - ▶ Forward masking has more significant effect (150-200ms; even relatively high-level sound is masked) than backward masking (5-10ms; only low-level sound can be masked)

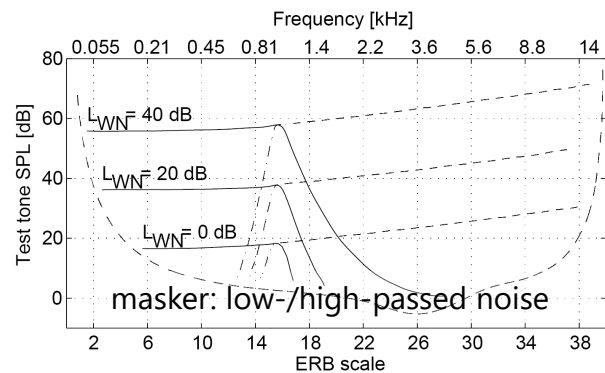
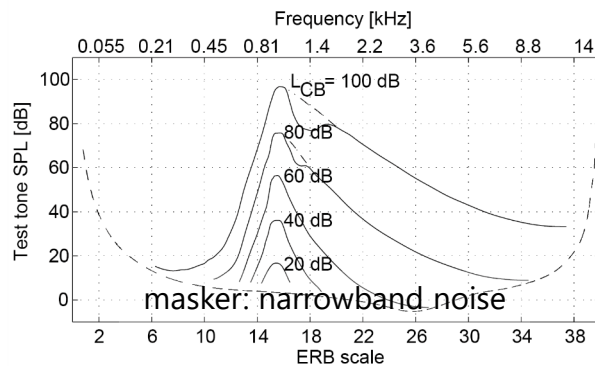
Spectral masking

- ▶ Effect of spectral masking changes depending on the type of masker (i.e. noise).
- ▶ White noise masker raises the threshold of audibility across the whole frequency band.
- ▶ Narrowband (band-passed) noise masker raises the threshold just around the centre frequency of the band.



Spectral masking

- ▶ Increasing the level of narrowband noise masker also raises the threshold.
- ▶ Similar to narrowband noise, both low-pass or high-pass filtered noises also raise the threshold only around the frequency where the masker's energy is dominant.



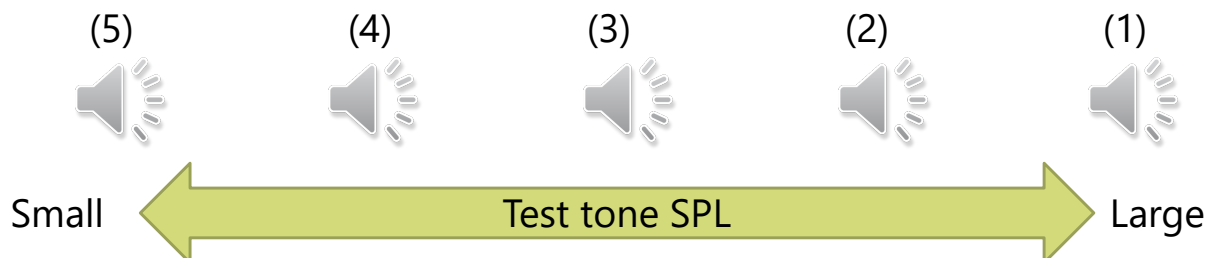
Pulkki, Fig.9.6&9.7

▶ 19

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Example

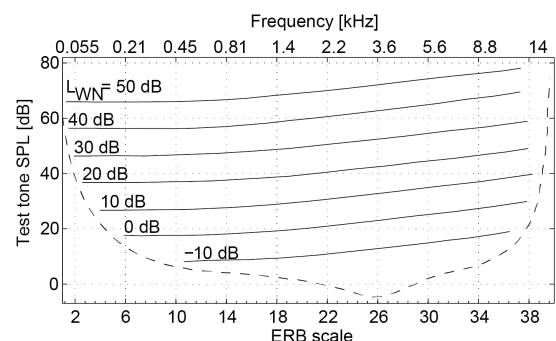
- ▶ Spectral masking by white noise



When playing from (1) to (5), down to which level is the tone audible?



<https://thehearinginstitute.org/hearing-clinic/audiology-services/hearing-tests/>

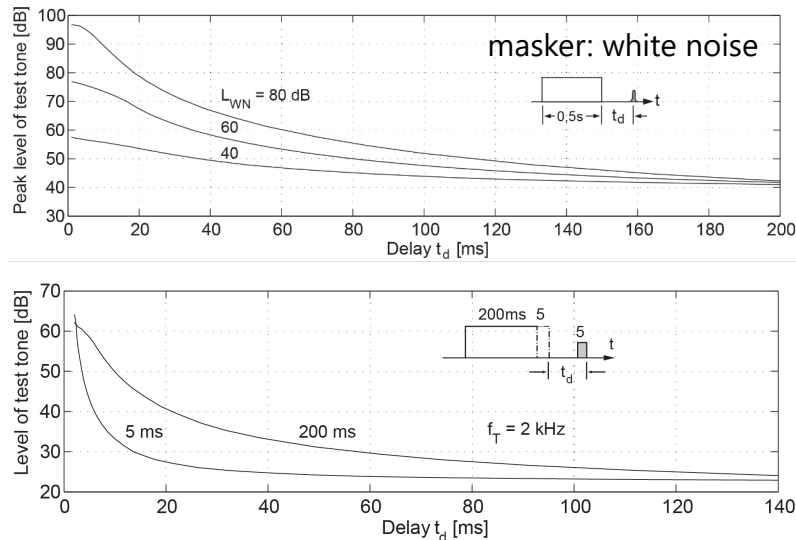


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Temporal masking

- ▶ For forward temporal masking, the threshold of hearing is affected by:
 - ▶ The level of masker
 - ▶ The temporal length of masker



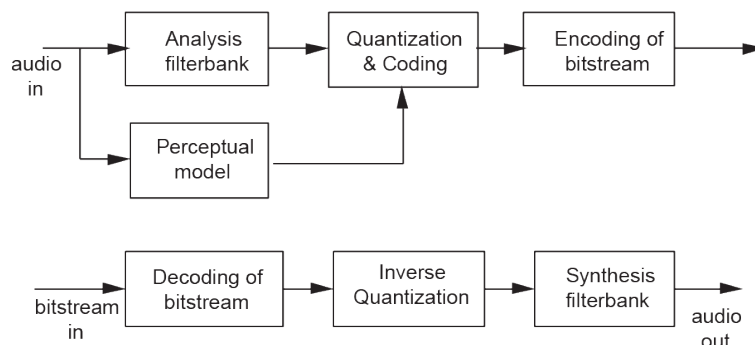
Pulkki, Fig.9.11&9.12

▶ 21

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Practical example

- ▶ Some *audio codec* (coding/decoding) use perceptual masking model to reduce the bit-rate (i.e. data size).
- ▶ The most commonly used (spectral) masking-based audio codec is MPEG-1 Layer-3 also known as **MP3**.
- ▶ The technique varies quantisation levels (i.e. bit rate) to shape the spectrum of the quantisation noise so that it follows the masking curve created by the target signal.

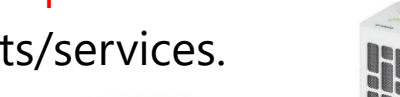


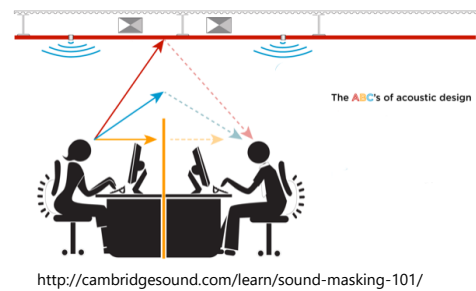
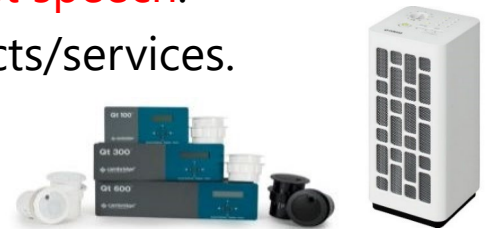
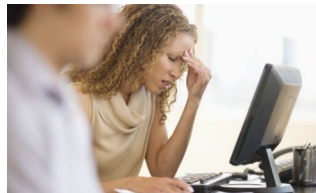
Pulkki, Fig.15.14

▶ 22

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Practical example

- ▶ **Sound masking system** is another practical example of technology making use of auditory masking.
 - ▶ A **masking sound** (jammer sound) is played to hinder understanding the context of **target speech**.
 - ▶ Used in various commercial products/services.
 - ▶ Examples of masking sound:
 - ▶ Pink noise
 - ▶ Babble noise
 - ▶ Speech-like noise [Hioka 2020]
- 
- The diagram illustrates a sound masking system. It shows a stack of three blue boxes labeled 'QS 100', 'QS 300', and 'QS 600'. To the right of the boxes are two white cylindrical speakers and one black cylindrical speaker. A red line with blue wavy arrows represents a sound wave or signal path. A red arrow points from the bottom of the stack of boxes to the red line. A white building is visible in the background on the right.

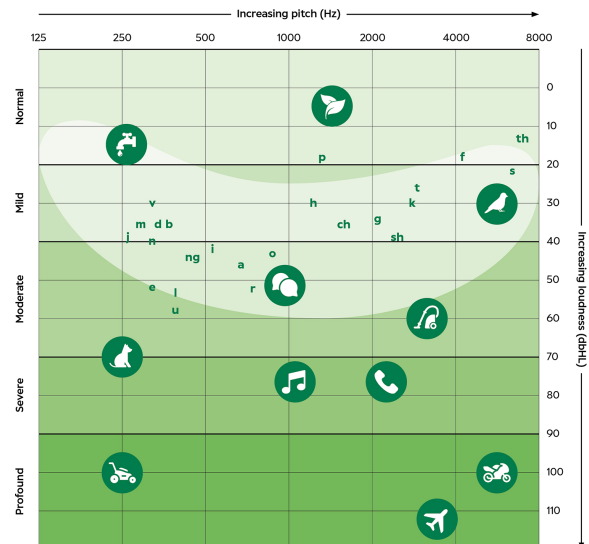


▶ 23

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Practical example

- Audiogram



<https://www.specsavers.co.nz/hearing/understanding-an-audiogram?srsId=AfmBOoZoMQ8cZSwNypqjVusl8HF7J17l0hgKI3WcaNlRsn6cXDIX8l>

<https://www.auckland.ac.nz/en/on-campus/facilities-and-services/university-of-auckland-clinics/audiology-teaching-clinics/hearing-appointments.html>

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Section 3: Psychology of hearing

25

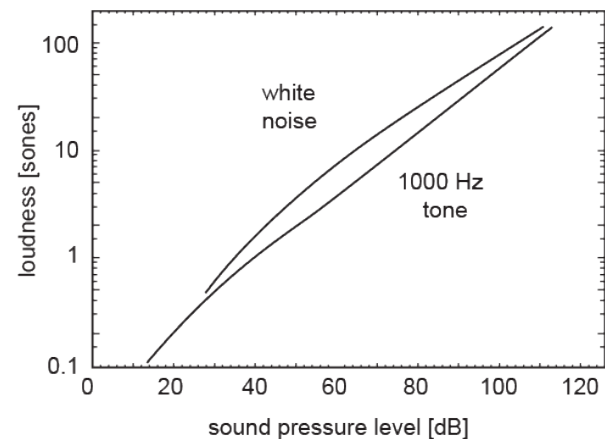
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Physical and psychoacoustic quantities

- ▶ So far we have learnt that the basic functioning of hearing resembles a band-pass filterbank
- ▶ For an objective sound, this process characterises a sound by their:
 - ▶ Frequency
 - ▶ Amplitude
 - ▶ Amplitude spectrum
 - ▶ Temporal length
- ▶ For a subjective sound, these correspond to:
 - ▶ Pitch
 - ▶ Loudness
 - ▶ Timbre
 - ▶ Duration

Loudness

- ▶ **Loudness** represents auditory sensation in terms of which sounds can be ordered from quiet to loud
- ▶ Determined by psychological test where subject finds the amplitude of sound that is "twice as loud" as a reference
- ▶ Unit: **sone**, 1 sone = 1 kHz tone at 40 dB SPL
- ▶ +10 dB SPL ~ x 2 sone
- ▶ Loudness increases when the sound has a broadband amplitude spectrum, e.g. white noise > 1 kHz tone



▶ 27

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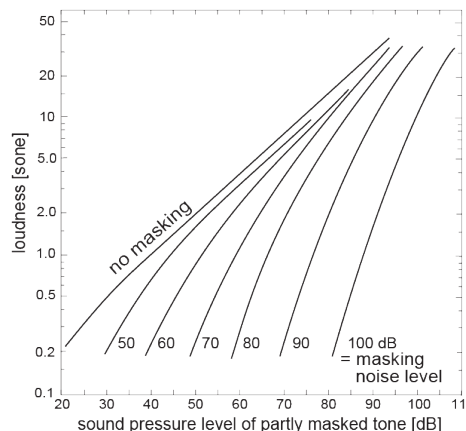
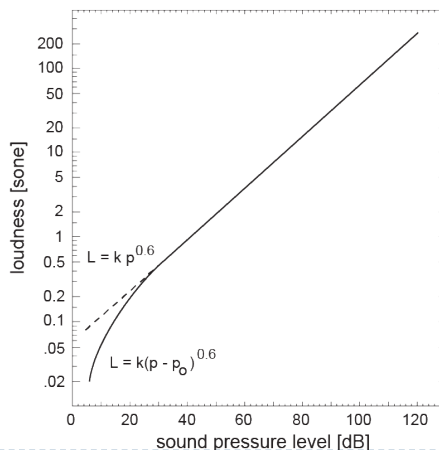
Pulkki, Fig.10.7

Loudness

- ▶ Loudness-SPL relationship can be expressed by simple exponential function

$$L = kp^{0.6} \quad p > 40 \text{ dB SPL}$$

$$L = k(p - p_0)^{0.6} \quad p < 40 \text{ dB SPL}$$
- ▶ Loudness is affected when the target is masked by noise. The higher the noise level, the lower the target's loudness



Pulkki, Fig.10.9&10.10

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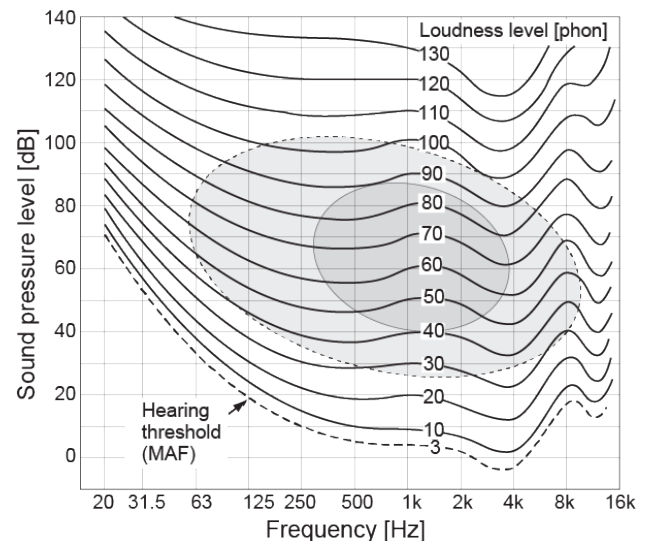
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Loudness

- ▶ **Equal loudness curve** can also be used to represent loudness, but this is called **loudness level**
- ▶ Unit: **phon**, 1 phon = as loud as 1 dB SPL at 1 kHz
- ▶ +10 dB SPL ~ + 10 phon

Loudness level vs Loudness

- ▶ Different relationship with dB
 - ▶ Sone: log scale progression
 - ▶ Phon: linear scale progression



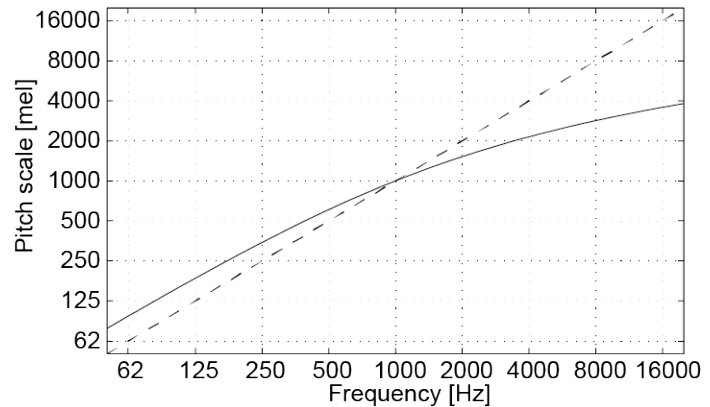
Pitch

- ▶ **Pitch** is defined as "*auditory attribute of sound according to which sounds can be ordered on a scale from low to high*"
- ▶ Can be perceived from various types of sound:
 - ▶ sinusoidal signal (alarm, sign)
 - ▶ vocals (gender of speaker, prosody)
 - ▶ instrument sounds (melody)
 - ▶ noise (sound produced by objects)
- ▶ A few **scales** to quantify perception of pitch have been proposed some of which are based on results of psychological experiments.

Pitch

- ▶ *Mel scale* represents average human perception of pitch against frequency of sound.
- ▶ It suggests doubling frequency (i.e. 1 octave) of sound:
 - ▶ < 500 Hz feels approximately "two times higher"
 - ▶ > 500 Hz feels much lower than "two times higher"
- ▶ Anchor point is 1 kHz = 1000 mel
- ▶ Formulated by

$$\text{mel} = 2595 \log_{10}(1 + f / 700)$$



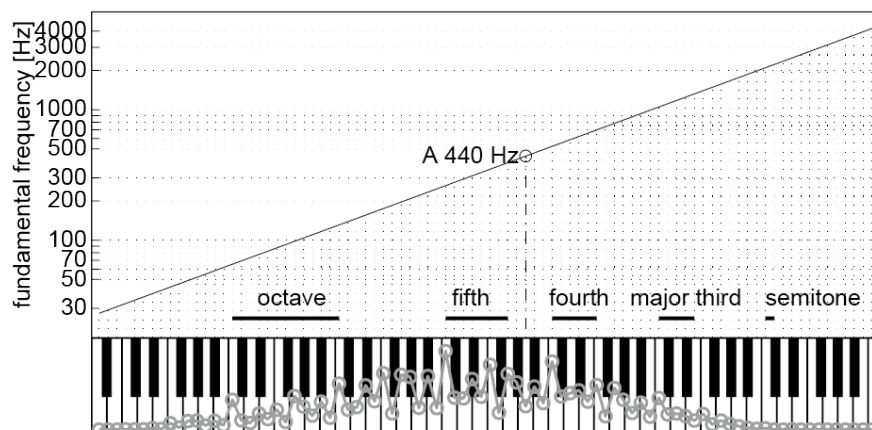
▶ 31

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Pulkki, Fig.10.5

Pitch

- ▶ For musical notes, *logarithmic frequency scale* is often used to define their pitches.
- ▶ For example *equal temperament* divides an octave logarithmically into 12 semitones, making the interval (ratio of frequency) between adjacent notes to be $2^{\frac{1}{12}}$.



Pulkki, Fig.10.6

▶ 32

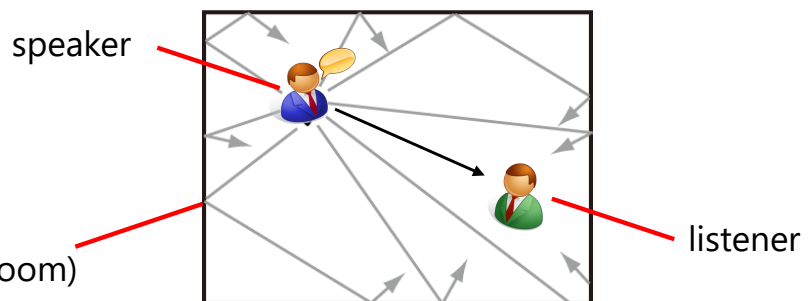
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Speech intelligibility

- ▶ **Speech intelligibility** (or simply intelligibility) is a property referring to how well the meaning of a spoken message is transmitted to a listener.
- ▶ Intelligibility depends on:
 - ▶ Ability of the **speaker** to produce acoustically and linguistically clear message.
 - ▶ How well the **transmission channel** is able to transmit the message.
 - ▶ How well the **listener** is able to receive and analyse the message.

e.g. conversation in a room

Transmission channel (sound propagation in the room)



▶ 33

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Speech intelligibility

- ▶ Another factor that affects speech intelligibility is the existence of interfering sound (noise) in the environment.
- ▶ The interfering sound causes auditory masking which makes the target speech less intelligible.
- ▶ Intelligibility is further reduced when it is compounded with the other factors discussed.

e.g. speech intelligibility under masking sound is significantly reduced when the listener is a non-native speaker.

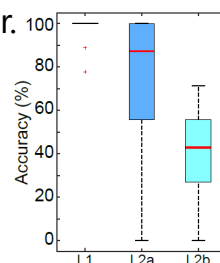


Figure 2: Accuracy rates in condition without masker (%).

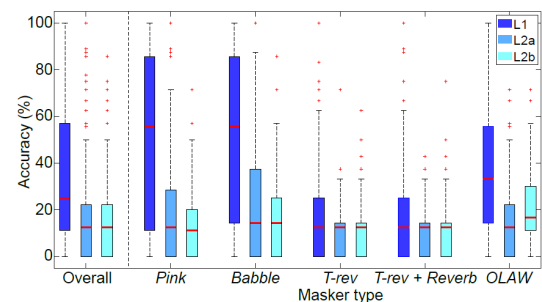


Figure 3: Accuracy rates in conditions with maskers (%).

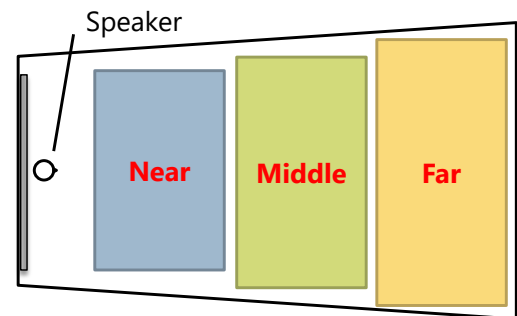
Masuda, Hioka *et al.* 2019

▶ 34

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Experiment

- ▶ A volunteer needed!
- ▶ Experimental procedures:
 1. The volunteer reads sentences from the lectern but only once
 2. Students on the floor transcribe what the volunteer reads for each sentence through the following link:
"Module>In-class experiments" on Canvas.
 3. We check the answer after several sets.



▶ 35

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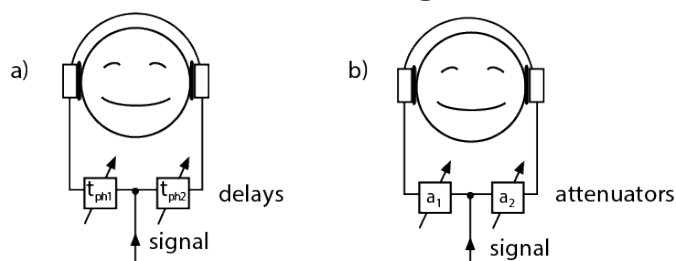
Section 4: Spatial hearing

Spatial hearing

- ▶ **Spatial hearing** refers to human's ability to **localise** sound source using our (two) ears.
- ▶ Localisation can be described by the perception of the source's **direction**, **distance**, and **spatial extent**.
- ▶ Two concepts used in spatial hearing.
 - ▶ **Monaural hearing**: listening in conditions where there is no interaural difference information, e.g. same signal enters both ears.
 - ▶ **Binaural hearing**: listening to sounds where information due to interaural differences exists and is taken into account.
- ▶ More information about the acoustical environment (acoustic cues) is provided in binaural hearing.

Binaural hearing

- ▶ Binaural cues are derived from the difference in the signals between the two ears. There are two main acoustic cues:
 - ▶ **Interaural time difference (ITD)** occurs due to difference of distance for a sound to propagate from the source to the ears.
 - ▶ **Interaural level difference (ILD)** occurs due to the scattering (shadowing, reflection) of sound waves by the head and torso.
- ▶ Binaural hearing can be offered by projecting sound with controlled ITD/ILD through headphones.



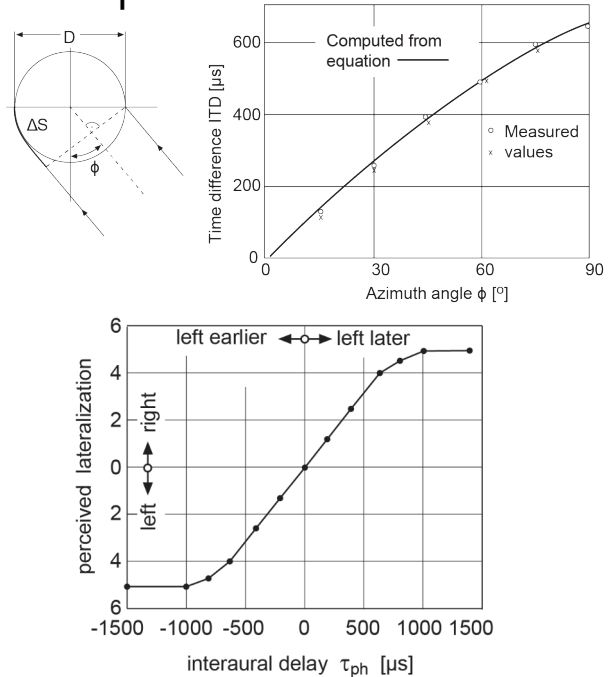
Pulkki, Fig.12.1

Binaural hearing

- ▶ ITD can be approximated by spherical head model where the value varies from 0 to ~600-700 μs .

$$\tau = \frac{D}{2c}(\varphi + \sin\varphi)$$

- ▶ More distinctive spatial cue in lower frequency band (< 1.5 kHz)



▶ 39

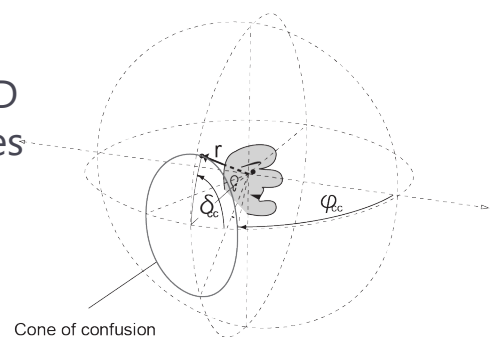
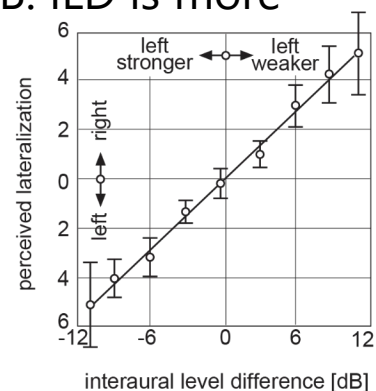
MECHENG726 2026 Chap3 Science of Hearing

Pulkki, Fig.12.9&12.11

Binaural hearing

- ▶ The range of ILD spans from -12 to 12 dB. ILD is more distinctive in higher frequency band
- ▶ Relying just on ILD/ITD causes the ambiguity known as *corn of confusion*. Additional cues are used to solve this ambiguity.

- ▶ **Spectral cues:** each ear uses spectral change of sound caused by the shape of head including pinna.
- ▶ **Dynamic cues:** inducing changes of ITD/ILD by rotating the head provide additional cues of source direction.



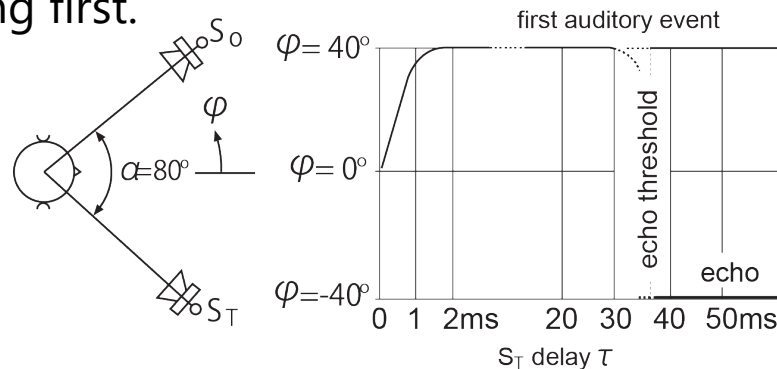
▶ 40

MECHENG726 2026 Chap3 Science of Hearing

Pulkki, Fig.12.3&12.11

Binaural hearing

- ▶ **Precedence effect** is an assisting mechanism of spatial hearing, which suppresses the effect of reflections from walls, ceiling and floor in enclosed spaces that arrive always after the direct sound.
- ▶ When sounds are arriving from two different angles with small delays, the localisation is affected by the sound arriving first.



Pulkki, Fig.12.20

Experiment

We do this experiment by splitting the class into two groups:

Group A

- ▶ Test1: Close your eyes and listen to the sound I play from my mobile phone. Point where I'm standing.
- ▶ Test2: Block one of your ears, then repeat Test1.

Group B

- ▶ Observe how students in Group A respond to each test.

Swap over between Group A and B then repeat the tests.